



Inventory decisions for a finite horizon problem with product substitution options and time varying demand



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ABSTRACT

The inventory control of substitutable products has been recognized as a problem worthy of study in the operations management literature. Product substitution provides flexibility in supply chain management and enhances response time in production control. This paper proposes a finite horizon inventory control problem for two substitutable products, which are ordered jointly in each replenishment epoch. Demand for the products are assumed to be time-varying. In case of a stock-out for one of the products, its demand is satisfied by using the stock of the other product. The optimal ordering schedule, for both products, that minimizes the total cost over a finite planning horizon is derived. Numerical examples along with sensitivity analyses are also presented.

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1. Introduction

It is well known that effective supply management concerns the ability to match production to demand in order to deal with uncertainty and variability. So flexibility tools are important in supply chain members (manufacturers, retailers and customers) as they permit adaptability and timely responses (Iravani et al. [1]). These flexibility tools help improve product availability, which has a positive impact on customer loyalty and the business performance of retailers and their suppliers/manufacturers, and can be exemplified by product substitution (Trautrimis et al. [2]).

Substitution is the practice of using an alternative product in order to meet the demand for another product. This practice has recently gained considerable attention in the operations management literature because it impacts the effectiveness of companies decisions regarding material/product planning, pricing and control (Shin et al. [3]). In general, substitution can take place at the end product level or at the component level and, according to Tang and Yin [4], there are three substitution mechanisms: inventory-based (stock-out), assortment-based and price-based substitution. Inventory-based substitution is observed in stock-out situations where it is common practice for consumers, who intend to purchase a particular product which is not available in the store, to be willing to buy a substitute product rather than visit a different shop. A survey mentions that 12%–18% of consumers said that they would not buy a product on a shopping trip if their favorite brand-size was not available; the rest indicated that they would be willing to buy another size of the same brand, or switch brands (Anupindi et al. [5]). In assortment-based substitution the customer chooses a substitute, triggered by the fact that the

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substitute is newly added in the assortment. In price-based substitution the customers substitution behavior is driven by a change in the relative price of the substitutable product. Alternative classifications of inventory models with substitutable products are found in Mahajan and van Ryzin [6], Tan and Karabati [7], and Stavroulaki [8].

The earliest EOQ model for inventory with substitution was examined in McGillivray and Silver [9]. Drezner et al. [10] considered an EOQ type model for two substitutable products and derived closed-form formulas for the optimal order quantities. They also compared the cases of full substitution and no substitution. Goyal [11] proposed adding a fixed (penalty) cost for substitution to the basic model of Drezner et al. [10] in order to allow for the possibility of full substitution, which is absent in the basic model of [10]. Gurnani and Drezner [12] extended the model of [10] to the case of multiple substitutable products and proposed a reformulation, based on the idea of run-out times, to determine the optimal order and substitution quantities. Chand et al. [13] studied a problem with substitution options in a manufacturing environment. They assumed that some components or parts may be used as substitutes for others in an assembly process. For this problem they derived the optimal order quantities of the parts based on an EOQ-type optimization model. Hsu et al. [14] considered a multi-product dynamic lot sizing problem where the products are indexed such that a lower-index product may be used to substitute for a higher-index product. The authors developed a dynamic programming algorithm to solve their problem. Recently, Salameh et al. [15] studied a two product joint replenishment model with substitution similar to that in [10], but allowing for partial substitution when stock-out occurs in one of the products. Maddah et al. [16] extended the model in [15] to multiple products. Krommyda et al. [17] studied a two substitutable products inventory problem under partial substitution where the demand of each product is a deterministic function of both stock levels. Ghoniem and Maddah [18] developed an integrated model for optimizing intertwined retail decisions on assortment, pricing and inventory for a product category composed of substitutable products, with a focus on fast-moving consumer goods that are characterized by deterministic demand patterns. The authors modeled the problem as a mixed-integer nonlinear program and proposed a linear reformulation, to obtain exact solutions to large instances of the problem in reasonable time. Based on an empirical analysis of real data, they showed that a significant portion of the demand variability is explained by the price variation and the substitution. For more information on general inventory and production models with product substitution: see Lang [19] and Shin et al. [3].

It is worth noting that deterministic inventory models with a finite planning horizon, time-varying demand rates and substitution are scarce. This scarcity is not shared by Economic Lot sizing models with time-varying demand rate which have been known for some time in the literature. See for example Resh et al. [20], Donaldson [21], Barbosa and Friedman [22], Henery [23,24], and Massonnet et al. [25]. Needless to say that the assumption of constant demand, besides its simplicity, is useful when supply contracts are signed ahead of time, designating deliveries for subsequent periods (see Bramel and Simchi-Levi [26]). However, in cases where the demand rate varies with time, the use of demand information over a finite planning period is required for the design of adequate inventory control policies. Moreover, inventory models with time varying demand cover a broad range of practical situations including multi-echelon assembly operations, production to contract, products with seasonal demand etc (see Silver et al. [27]). Hence the need for the inclusion of the time factor in demand in inventory models with substitution.

The present paper addresses the problem of inventory replenishment decisions for two substitutable products, continuous time-varying demand and a finite planning horizon. The products are ordered jointly in every replenishment cycle and have different demand rates depending on time. In case of a stock-out for one of the products, its demand is satisfied by using the stock of the other product. Product substitution affects the inventory control policy as it is not necessary to plan sufficient inventory for each product if the demands are totally independent since the inventory of the substitutable product can be used to satisfy the unmet demand and thus avoid lost sales (see Song [28]). The search for an optimal inventory policy that minimizes the total cost over a finite planning horizon is undertaken. It is shown that an optimal policy exists under some technical conditions.

The proposed model can be applied, as in [10], to products, where only one of them substitutes for the other (known as “one-way substitution”). One-way substitution occurs in many settings, such as in a manufacturing environment (substituting a lower-quality component or part with a higher-quality component or part), for example in the fabrication of semiconductor chips where a faster processor can be substituted for a slower processor (Hsu and Bassok, [29]). Also, this model could be used in a supplier-driven substitution framework where the supplier uses dual sourcing to meet the demand. Suggestions on possible applications of the model to two-way substitution are provided in the conclusion section.

The remainder of this paper is organized as follows. In Section 2, we discuss the assumptions for the proposed inventory model and the notation to be used throughout the paper. In Section 3, the mathematical model is developed. Details of the optimal replenishment policy are also presented. Numerical examples are provided for illustration in Section 4 and, finally, a summary and conclusions are found in Section 5. Proofs of the main technical results are contained in the appendices.

2. Assumptions and notations

The mathematical model is developed under the following assumptions:

1. A two product inventory system is considered over a known and finite planning horizon, $T > 0$.
2. The demand rate for product i ($i = 1, 2$) at time t is denoted by the function $D_i(t) : [0, T] \rightarrow \mathbb{R}_+$. It is assumed $D_i(t) \in C^1([0, T])$, where $C^1([0, T])$ is the space of differentiable functions on the interval $[0, T]$.

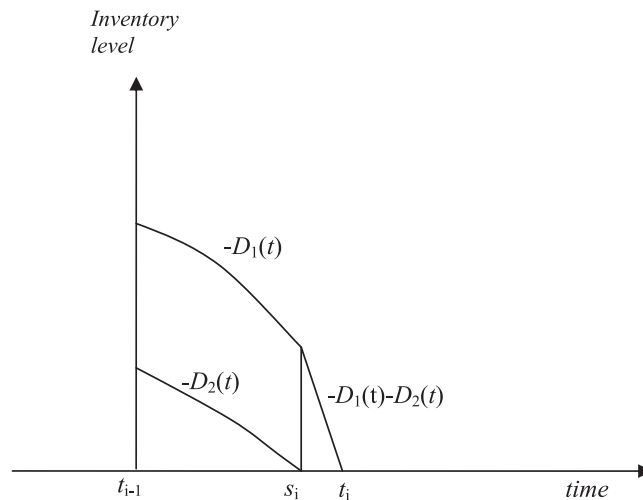


Fig. 1. The inventory levels for the two products during the cycle i

3. During a cycle, both products face their own demand. One of the products, say product 2, exhausts first (this happens in each cycle due to, for example, better color or flavor, etc) and product 1 is used to satisfy its demand (say sequence $2 \rightarrow 1$). The cycle starts with an instant replenishment of both products and ends when the inventory of product 1 reaches level zero.
4. Shortages are not allowed.

The notation used throughout the paper is presented below:

k	The set-up cost, $k > 0$
$D_i(t)$	The demand for product i at time t , $i = 1, 2$.
h_i	The holding costs for product i , ($h_i > 0$) $i = 1, 2$, with $h_1 \neq h_2$
c	The transfer cost, $c > 0$.
t_{i-1}	The starting time of cycle i .
t_i	The ending time of cycle i .
s_i	The time of substitution in cycle i .
$I_i(t)$	The inventory level for product i , $i = 1, 2$ at time t .
$'$	The derivative of a univariate function.
∂_x	The partial derivative of a bivariate function with respect to the first variable.
∂_y	The partial derivative of a bivariate function with respect to the second variable.
∂_x^2	The second derivative of a bivariate function with respect to the first variable.
∂_y^2	The second derivative of a bivariate function with respect to the second variable.
$\partial_x \partial_y$	The cross partial derivative of a bivariate function.
∇	The gradient of a bivariate function.

3. The mathematical formulation of the problem

Fig. 1 depicts the inventory of two substitutable products with respect to time in a typical cycle $[t_{i-1}, t_i]$ with time-varying demand rates. At time instant s_i the inventory of product 2 depletes and its demand is subsequently satisfied by the inventory of product 1. At time instant t_i the inventory level of product 1 drops to zero and an instant joint replenishment for both products is triggered.

So, on the time interval $(t_{i-1}, s_i]$ the change in inventory level for products 1 and 2 is described by the Eqs. (1) and (2), respectively:

$$\frac{dI_1(t)}{dt} = -D_1(t), \quad t_{i-1} < t \leq s_i, \quad (1)$$

with $I_1(s_i^-) = I_1(s_i^+)$. Also,

$$\frac{dI_2(t)}{dt} = -D_2(t), \quad t_{i-1} < t \leq s_i, \quad (2)$$

with $I_2(s_i) = 0$, while during the time interval $[s_i, t_i]$ the variation of the inventory level of product 1 is described as:

$$\frac{dI_1(t)}{dt} = -D_1(t) - D_2(t), \quad s_i \leq t < t_i, \quad (3)$$

with $I_1(t_i^-) = 0$. The solutions of the above differential equations are:

$$I_1(t) = \int_t^{t_i} D_1(u)du + \int_{s_i}^{t_i} D_2(u)du, \quad t_{i-1} < t \leq s_i, \quad (4)$$

$$I_2(t) = \int_t^{s_i} D_2(u)du, \quad t_{i-1} < t \leq s_i, \quad (5)$$

$$I_1(t) = \int_t^{t_i} (D_1(u) + D_2(u))du, \quad s_i \leq t < t_i. \quad (6)$$

Direct computations show that the holding cost for product 1 is:

$$H_1 := h_1 \left[\int_{t_{i-1}}^{s_i} (t - t_{i-1}) D_1(t) dt + \int_{s_i}^{t_i} (t - t_{i-1}) \{D_1(t) + D_2(t)\} dt \right], \quad (7)$$

the holding cost for product 2 is:

$$H_2 := h_2 \int_{t_{i-1}}^{s_i} (t - t_{i-1}) D_2(t) dt, \quad (8)$$

and the transfer cost is:

$$H_3 := c \int_{s_i}^{t_i} D_2(t) dt. \quad (9)$$

It can be shown that: $H_1 + H_2 + H_3$ is equal to:

$$\int_{t_{i-1}}^{t_i} \{h_1 D_1(t) + h_2 D_2(t)\} (t - t_{i-1}) dt + \int_{s_i}^{t_i} \{c - (h_2 - h_1)(t - t_{i-1})\} D_2(t) dt. \quad (10)$$

For a given cycle starting at time x and ending at time y with substitution time s ($x \leq s \leq y$), define the sum of holding and transfer costs as:

$$V(x, s, y) := \int_x^y \{h_1 D_1(t) + h_2 D_2(t)\} (t - x) dt + \int_s^y \{c - (h_2 - h_1)(t - x)\} D_2(t) dt. \quad (11)$$

The problem of determining the optimal inventory policy is formulated as the following mixed integer nonlinear program (MINLP):

$$\begin{aligned} P1 : \min z &= nk + \sum_{i=1}^n V(t_{i-1}, s_i, t_i) \\ \text{subject to : } 0 &= t_0 \leq t_{i-1} \leq \dots \leq t_n = T. \\ t_{i-1} &\leq s_i \leq t_i, \quad i = 1, \dots, n. \end{aligned} \quad (12)$$

The optimal inventory policy reduces to finding n and the vectors (t_{i-1}, s_i, t_i) , $i = 1, \dots, n$ which minimizes z .

It is clear that, when the optimal values for n , t_i , s_i , $i = 1, \dots, n$ (say n^* , t_i^* , s_i^* , $i = 1 \dots n^*$) are determined, the optimal ordering quantities for the two products can be determined: $I_1(t_i^*)$, $i = 1 \dots n^*$ (ordering quantity for product 1 at replenishment instant t_i^*), $I_2(t_i^*)$, $i = 1 \dots n^*$ (ordering quantity for product 2 at replenishment instant t_i^*).

Note that:

(i) If $s = y$, then this leads to the no substitution case and

$$V(x, s, y) := \int_x^y \{h_1 D_1(t) + h_2 D_2(t)\} (t - x) dt. \quad (13)$$

(ii) If $s = x$, then this leads to the full substitution case and

$$V(x, s, y) := h_1 \int_x^y (t - x) \{D_1(t) + D_2(t)\} dt + c \int_x^y D_2(t) dt. \quad (14)$$

(iii) If $x < s < y$, then we have partial substitution.

The next result is related to the possibility that cases (i)–(iii) occur. Let $\tilde{P}1$ the nonlinear program obtained from $P1$ by fixing the integer variable n .

Theorem 1. If Problem $\tilde{P}1$ has a solution $(\hat{t}_{i-1}, \hat{s}_i, \hat{t}_i)$ for $i = 1, \dots, n$, then

$$\hat{s}_i = \begin{cases} \hat{t}_i & \text{if } \gamma > \hat{t}_i - \hat{t}_{i-1} > 0 \text{ or } \gamma \leq 0 \\ \hat{t}_{i-1} + \gamma & \text{if } \hat{t}_{i-1} < \hat{t}_{i-1} + \gamma \leq \hat{t}_i, \end{cases} \quad (15)$$

where

$$\gamma = \frac{c}{h_2 - h_1}. \quad (16)$$

The proof of the theorem is found in [Appendix A](#).

Remark 1.

1. [Theorem 1](#) shows that, at the optimal solution, full substitution (case (ii)) does not occur. The same argument shows that it does not also occur in the EOQ model with constant demand. This was pointed out by Drezner et al. [10] and remarked on by Goyal [11]. We shall comment on this case later on: see [Remark 3](#) below.
2. The substitution is possible if $\gamma > 0$ and $c < (h_2 - h_1)(\hat{t}_i - \hat{t}_{i-1})$. The first relation implies that $h_2 > h_1$ and, consequently, it is reasonable to expect that product 2 is substituted for by product 1. However, the second relation requires that the profit from the substitution (in terms of holding cost) should be greater than the transfer cost. Otherwise no substitution occurs. This can also be seen by noticing that the second integral term on the right hand-side of (11) can be written as

$$-(h_2 - h_1) \int_s^y (t - s) D_2(t) dt + \{c - (h_2 - h_1)(s - x)\} \int_s^y D_2(t) dt, \quad (17)$$

so when $(h_2 - h_1) < 0$ no substitution (i.e. $s = y$) is more profitable.

3. The model can easily be modified so that a fraction of the demand for product 2 is satisfied by product 1. To this end, a cost per unit of lost demand could be taken into account, which can be handled in the same manner as the transfer cost.

Guided by the preceding analysis, in order to solve P1, we shall consider separately the cases $\gamma > 0$ (this is equivalent to $h_2 > h_1$) and $\gamma < 0$ (this is equivalent to $h_2 < h_1$). Note that when $\gamma > 0$ cases (i) and (iii) above may occur and when $\gamma < 0$ only case (i) occurs. So, since s is expressed as a function of x and y (i.e. $s = s(x, y)$), the sum of the holding and transfer costs $V(x, s, y)$ will be expressed as a function of x and y .

Let

$$\Omega = \{(x, y) : 0 < x < y \leq T\}. \quad (18)$$

For $j = 1, 2$, and $(x, y) \in \Omega$ define

$$r_j(x, y) = \int_x^y (t - x) D_j(t) dt, \quad (19)$$

$$R_1(x, y) := h_1 r_1(x, y). \quad (20)$$

For $\gamma > 0$, using (10) with (17) and [Theorem 1](#), let

$$R_2(x, y) := h_2 r_2(x, y) - (h_2 - h_1) \begin{cases} r_2(x + \gamma, y), & \text{if } x + \gamma \leq y \\ 0, & \text{otherwise} \end{cases}. \quad (21)$$

For $\gamma < 0$, using (10) with (17) and [Theorem 1](#), set

$$R_2(x, y) := h_2 r_2(x, y) \quad (22)$$

and consequently

$$V(x, s(x, y), y) = R(x, y) := R_1(x, y) + R_2(x, y). \quad (23)$$

3.1. The case $\gamma > 0$.

Using [Theorem 1](#) Problem P1 defined in (12) is equivalent to the following MINLP:

$$\text{P2. min } z = nk + \sum_{i=1}^n R(t_{i-1}, t_i)$$

$$\text{subject to : } 0 = t_0 \leq t_{i-1} \leq \dots \leq t_n = T.$$

where R is given by (23) and the decision variables are: n and t_i , $i = 1, \dots, n - 1$.

[Theorems 2](#) and [3](#) below are the main results of the paper. [Theorem 2](#) states that for fixed n , P2 has a unique solution under some technical conditions. [Theorem 3](#) proves the convexity of the objective function with respect to n at optimal t_i^* 's, $i = 1, \dots, n - 1$. This, as it turned out, is crucial to obtaining the optimal inventory policy. Problem P2 as it stands is new in the literature for inventory models with substitutions. If $h_1 = 0$, then P2 has some technical similarities to a problem treated in Benkherouf and Gilding [30] concerning optimal replenishment policies for a finite planning horizon inventory model with deteriorating items and a permissible delay in payments. To solve P2 we shall draw on ideas from Benkherouf and Gilding ([31] and [30]).

3.1.1. Preliminaries

The next lemma shows that the functions R_1 and R_2 defined in (20) and (21) respectively are differentiable on the set $\bar{\Omega}$, where $\bar{\Omega}$ is the closure of the set Ω .

Lemma 1. *The functions R_1 , and $R_2 \in C^1(\bar{\Omega})$.*

The proof of the Lemma is found in Appendix B. Before we proceed further, the following computations are needed. For $j = 1, 2$

$$\partial_x r_j(x, y) = - \int_x^y D_j(t) dt, \quad (24)$$

$$\partial_y r_j(x, y) = (y - x) D_j(y), \quad (25)$$

$$\partial_x \partial_y r_j(x, y) = -D_j(y), \quad (26)$$

$$\partial_x^2 r_j(x, y) = D_j(x), \quad (27)$$

$$\partial_y^2 r_j(x, y) = D_j(y) + (y - x) D'_j(y). \quad (28)$$

Remark 2. Note that the function R in (23) is not twice differentiable in $\bar{\Omega}$.

Indeed, for $j = 1, 2$, the function $r_j(x, y)$ is twice differentiable in $\bar{\Omega}$. If differentiability problems occur then these happen on the line segment $(x, x + \gamma)$. We have by (26)–(28)

$$\partial_x^2 r_2(x, x + \gamma) = D_2(x), \quad (29)$$

$$\partial_y^2 r_2(x, x + \gamma) = D_2(x + \gamma) + \gamma D'_2(x + \gamma), \quad (30)$$

$$\partial_x \partial_y r_2(x, x + \gamma) = -D_2(x + \gamma). \quad (31)$$

For $x + \gamma < y$, $\partial_x^2 r_2(x, x + \gamma) \neq 0$, $\partial_y^2 r_1(x, x + \gamma) \neq 0$, and $\partial_x \partial_y r_1(x, x + \gamma) \neq 0$. Therefore, R is not twice differentiable by (21).

Our objective next is to solve P2 by appealing to the established methodology in Benkherouf and Gilding [31]. This methodology requires that R be twice differentiable. Fortunately, as we shall see, this will not preclude its applicability to P2. Special care needs to be taken at the points where R fails to be second order differentiable.

For an arbitrary function Z , let $(\partial_x Z)(x^+, y)$, $(\partial_y Z)(x, y^+)$ denote the partial derivative of Z with respect to x and y from the right, respectively. Also, let $(\partial_x Z)(x^-, y)$, $(\partial_y Z)(x, y^-)$ denote the partial derivative of Z with respect to x and y from the left, respectively. A key requirement for the methodology developed in [31] to be applicable is $\partial_x \partial_y R(x, y) < 0$ in Ω . However, the definition of R in (23) with the presence of two demand rate functions (D_1 and D_2) means that $\partial_x \partial_y R_1(x, y) < 0$, $\partial_x \partial_y R_2(x^\pm, y) < 0$, and $\partial_x \partial_y R_2(x, y^\pm) < 0$ in Ω will suffice. Here, $\partial_x \partial_y R_2(x^\pm, y)$ stands for cross partial derivatives on the right and left for the first argument. $\partial_x \partial_y R_2(x, y^\pm)$ refers to the cross partial derivatives on the right and the left for the second argument. The requirement $\partial_x \partial_y R_1(x, y) < 0$ is trivial from (26) and the definition of R_1 in (20). For R_2 , we shall check the requirement on the sets

$$\Omega_1 = \{(x, y) : 0 < x < y, \text{ and } x + \gamma < y\}, \quad (32)$$

$$\Omega_2 = \{(x, y) : 0 < x < y, \text{ and } x + \gamma > y\}. \quad (33)$$

In Ω_2 , the definition of R_2 in (21) and (26) gives $\partial_x \partial_y R_2(x, y) = -h_2 D_2(y)$, which is strictly negative. In Ω_1 , $\partial_x \partial_y R_2(x, y) = -h_1 D_2(y)$, which is strictly negative. This completes the proof.

As a result we have, for $(x, y) \in \Omega$,

$$\partial_x \partial_y R(x^\pm, y) < 0, \quad (34)$$

$$\partial_x \partial_y R(x, y^\pm) < 0. \quad (35)$$

Expressions (34) and (35) imply that the cost function R is submodular in Ω . For more information on submodular functions and their properties see Topkis [32].

For some functions z and f_j ($j = 1, 2$) define the operators

$$\mathcal{L}_x^{(j)} z := \partial_x^2 z + \partial_x \partial_y z + f_j(x) \partial_x z, \quad (36)$$

$$\mathcal{L}_y^{(j)} z := \partial_y^2 z + \partial_x \partial_y z + f_j(y) \partial_y z. \quad (37)$$

Applications of the operators (36) and (37) in the literature have been for the case $j = 1$. The present paper appears to be the first to call for application of these operators beyond $j = 1$.

Lemma 2. For $j = 1, 2$, let

$$f_j(t) := -\frac{D'_j(t)}{D_j(t)}, \quad (38)$$

then the functions $\mathcal{L}_x^{(j)} R_j$, and $\mathcal{L}_y^{(j)} R_j$ are continuous in $\bar{\Omega}$. Moreover, if f_j is non-decreasing, then $\mathcal{L}_x^{(j)} R_j \geq 0$ and $\mathcal{L}_y^{(j)} R_j = 0$, in $\bar{\Omega}$.

The proof of the Lemma is found in [Appendix C](#).

3.1.2. Some basic assumptions

Before we proceed further, we shall make the following assumptions:

Assumptions

(A1) For $j = 1, 2$, $f_1(t) = f_2(t) = f(t)$.

(A2) The function f is non-decreasing in t .

For the above assumptions, we note that:

1. Assumption (A1), implies that $D_1(t) = AD_2(t)$, where A is some strictly positive constant. This means that the demand for product 1 is proportional to the demand for product 2. From a practical point of view this assumption is met in the Joint Replenishment Problem (JRP), which concerns the coordinated replenishment of a group of items (in the same category) that may be jointly ordered from a single supplier (see [\[15\]](#)).
2. Assumption (A1) ensures that Problem P2 is solvable using the methodology of Benkherouf and Gilding [\[31\]](#).
3. Assumption (A2) reduces to the usual condition that the demand functions D_1 and D_2 are log-concave. This assumption is also found in Henery [\[23\]](#), where an economic lot size model is investigated.
4. Under Assumption (A1), and omitting the superscript notation, we may write

$$\mathcal{L}_x R = \mathcal{L}_x R_1 + \mathcal{L}_x R_2, \quad (39)$$

$$\mathcal{L}_y R = \mathcal{L}_y R_1 + \mathcal{L}_y R_2. \quad (40)$$

Also, for $(x, y) \in \Omega$, we have

$$\mathcal{L}_x R \geq 0, \quad \mathcal{L}_y R \geq 0, \quad (41)$$

under Assumption (A2).

5. The possibility that assumptions A1 and A2 fail to hold cannot be ruled out in applications. The implications of this possibility will be discussed later in this section.

3.1.3. The optimal inventory policy

This subsection contains the details of the optimal inventory policy for the finite horizon inventory model with substitution. As a matter of fact, a procedure for determining the optimal policy initially fixes the integer variable n in P2, resulting in a pure nonlinear programming problem, the solution to which is given by the following theorem.

Theorem 2. Under assumptions A1 and A2, and for fixed n , Problem P2 has a unique optimal solution. This solution is obtained by setting

$$\nabla G_n(t_0, t_1, \dots, t_n) = 0,$$

where

$$G_n(t_0, t_1, \dots, t_n) = \sum_{i=1}^n R(t_{i-1}, t_i), \quad (42)$$

where $t_0 = 0$, $t_n = T$, and R is given by (23).

The proof of the theorem is found in [Appendix D](#).

The following Theorem is concerned with the determination of the optimal number of replenishment cycles. The proof is omitted and may be found in Benkherouf and Gilding [\[31\]](#) and Denardo et al. [\[33\]](#). Write

$$g_n := g_n(T) = \min_{t_0, t_1, \dots, t_n} G_n(t_0, t_1, \dots, t_n),$$

for the optimal value of $G_n(t_0, t_1, \dots, t_n)$ for fixed n , where G_n is defined in (42).

Table 1
The optimal replenishment schedule for Example 1.

t_0^*	s_1^*	t_1^*	s_2^*	t_2^*
0	2.134	2.134	4.634	5

Theorem 3. The optimal number of cycles n^* in Problem P2 is given by: (i) if $k > g_1 - g_2$, then $n^* = 1$,
(ii) if there exists $N \geq 2$ such that $g_{N-1} - g_N > k > g_N - g_{N+1}$, then $n^* = N$,
(iii) if there exists $N \geq 2$ such that $k = g_N - g_{N+1}$, then $n^* = N$, and $n^* = N + 1$.

Theorem 3 shows that the function g_n is convex in n . This theorem ensures the existence of optimal n (note that $\lim_{n \rightarrow \infty} g_n = \infty$).

3.2. The case $\gamma < 0$.

For $j = 1, 2$ define

$$R_j(x, y) = c_j r_j(x, y), \quad (43)$$

where r_j is defined in (19).

The problem of finding an optimal inventory policy, P2, in the previous case, can then be restated with

$$R(x, y) = R_1(x, y) + R_2(x, y), \quad (44)$$

where R_j is defined in (43) for $j = 1, 2$.

The function R given by (44) is smooth in Ω . Therefore, the results for the case $\gamma < 0$ can be recovered under assumptions (A1) and (A2). Details are omitted since the set-up is similar to that for the case $\gamma > 0$. As a matter of fact the search for the optimal inventory policy is simpler than in the previous case since the singular points are absent.

Remark 3.

1. Note that if assumptions A1-A2 fail to hold, **Theorem 2** still holds but the uniqueness of the solution cannot be ensured, while **Theorem 3** remains valid. Also, it is worth noting that Assumption A1 ensures that Relation (D9) (in the appendix) holds, which is the key to the proof of **Theorem 2**. This relation leads to (D10) and (D11), which are essential to the rest of the proof of the **Theorem 2**.
2. In [11] it was suggested, among other things, that adding a fixed (penalty) cost for substitution in the basic model of Drezner et al. [10] would allow for the possibility of the full substitution case, which is absent in the basic model. This is possible to achieve in the present model. However, the resulting optimization presents a challenging problem since continuity is lost. We suspect that the results developed in [33] would be key to future progress on this problem.
3. The possibility of $h_1 = h_2$ cannot be ruled out in practice. It is an easy exercise to see that this results in the no substitution case in all cycles, where the optimality is found using the same analysis as in the case $\gamma < 0$. Also the same phenomena reappears if $c = 0$ and $h_2 < h_1$. On the other hand, if $c = 0$ and $h_2 > h_1$ then full substitution is observed. The optimal policy is found using, for example, the method found in [31].

4. Numerical examples for different demand configurations

In this section, we present some examples along with their sensitivity analysis to illustrate the applicability of **Theorems 2** and **3**. In order to examine the effect of time varying demand rate, we consider scenarios with exponential and linear demand rates. In addition, we assume constant demand rates for both product. To this end, the demand rates are approximated by the average demand per unit i.e. $D_1(t) = D_1 = \frac{\int_0^T D_1(t) dt}{T}$, $D_2(t) = D_2 = \frac{\int_0^T D_2(t) dt}{T}$. Moreover, the implication of varying a number of key parameters on the model is examined.

4.1. Exponential decreasing demand

It is well documented in the literature that the demand of spare parts for a range of products, such as agriculture products, decreases with time exponentially (see for example Hill et al. [34] and Wee [35]).

For the first example we assume that $k = 1000$, $h_1 = 3$, $h_2 = 5$, $c = 5$, $T = 5$, $D_1(t) = 80e^{-0.2t}$, $D_2(t) = 60e^{-0.2t}$.

Computations show that the optimal number of orders is $n = 2$ and the optimal cost is 3923.76. The optimal values of t_i , s_i $i = 1, \dots, n$ are found in **Table 1**. Although $h_1 < h_2$ no substitution occurs during the first cycle, probably because of the high transfer cost. Also note that the length of the cycles increases as the demand decreases with time.

Table 2 contains the results for the optimal replenishment policy using the constant approximation for the demand rates, i.e. $D_1(t) = D_1 = 50.57$, $D_2(t) = D_2 = 37.93$. The corresponding cost is 4133.41. From **Table 2**, it seems that this approximation affects the replenishment schedule (compare with **Table 1**). Under constant demand (no time demand variability), there

Table 2

The optimal replenishment schedule for Example 1 assuming a constant demand rate for the two products.

t_0^*	s_1^*	t_1^*	s_2^*	t_2^*
0	2.5	2.5	5	5

Table 3

The optimal replenishment schedule for modified data of Example 1.

t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*
0	0.5	0.658	1.158	1.371	1.871	2.148
s_4^* 2.648	t_4^* 3.001	s_5^* 3.502	t_5^* 3.945	s_6^* 4.445	t_6^* 5	.

Table 4

The optimal replenishment schedule for modified data of Example 1 (constant demand).

t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*
0	0.5	0.833	1.333	1.667	2.167	2.5
s_4^* 3	t_4^* 3.333	s_5^* 3.833	t_5^* 4.167	s_6^* 4.667	t_6^* 5	.

is no product substitution in the optimal replenishment policy. This leads to a total cost increase of around 5.4%, which clearly shows the effect of demand variability on the total cost.

Next, the ordering cost k and transfer cost c are reduced to the values $k = 100$ and $c = 1$ while the other parameters remain unchanged. For these modified data the optimal number of orders is $n = 6$ and the optimal cost is 1254.62. The optimal values of t_i , s_i $i = 1, \dots, n$ are shown in Table 3. As was expected, the number of orders increases and, because of the low transfer cost, substitution occurs in all cycles. Assuming constant demand rates, the number of orders is $n = 6$ and the optimal cost is 1285.85. The replenishment policy is presented in Table 4.

In order to evaluate the effect of changing parameter values on optimal replenishment policy, both for constant and time-varying demand, we perform a sensitivity analysis on all the cost parameters for the initial data. The results presented in Table 5 summarize the sensitivity analysis with respect to the cost parameters, where i (see column 1 of Table 6) stands for the initial data of Example 1. Table 6 gives the sensitivity analysis for Example 1 (again with initial data), assuming $D_1(t) = D_1$ and $D_2(t) = D_2$. The results and insights obtained from Tables 5 and 6 are discussed at the end of this section.

4.2. Linearly increasing demand

It is well documented in the literature that demand for new product increases linearly with time when it substitutes an existing product in most electronics, automobiles and seasonal products with short lifetimes (see for example San Jose et al. [36]). We assume $D_1(t) = 96 + 6t$ and $D_2(t) = 80 + 5t$, while the cost parameters are as in Example 1. The optimal number of cycles is $n = 3$ and the optimal cost is 6360.06. Table 7 gives the optimal values of t_i , s_i $i = 1, \dots, n$.

Table 8 presents the optimal replenishment schedule assuming constant demand for the two products (i.e. $D_1(t) = D_1 = 111$ and $D_2(t) = D_2 = 92.5$) and the corresponding optimal cost is 6314.58. The optimal replenishment schedule indicates that no substitution occurs although $h_2 > h_1$. This is possibly due to the high value of the transfer cost coupled with the increasing nature (in time) of demand rates for the two products.

Next, the ordering cost k and transfer cost c are reduced to the values of $k = 100$ and $c = 1$ while the other data remain unchanged. For these modified data the optimal number of orders is $n = 10$ and the optimal cost is 1997.18. The optimal values of t_i , s_i $i = 1, \dots, n$ are shown in Table 9. Also, the length of cycles decreases as the demand increases. Moreover, substitution occurs in all cycles perhaps because of the low transfer cost. By assuming constant approximation for the demand rates the optimal number of orders is $n = 10$ and the optimal cost is 1994.37, while the optimal replenishment policy is presented in Table 10.

Table 11 presents the results of a sensitivity analysis with respect to some important model parameters, where i in the table refers to initial data of Example 2. A sensitivity analysis for the constant demand rate approximations is presented in Table 12.

The main managerial insights from the above two examples, and especially from Tables 5–6 and 11–12, are:

1. Tables 3, 4 and 7, 8 show that the use of a decreasing-in-time exponential function and an increasing-in-time linear function to describe the demand rate affects the replenishment schedule significantly. In addition, Tables 5 and 6 and 11 and 12 indicate that differences in the replenishment schedules and costs are notable when a constant approximation for the demand rate is used. This illustrates the need to use an accurate demand forecast. Moreover,

Table 5
Sensitivity analysis for Example 1.

parameter (i)	n^*	optimal policy								Optimal cost
$k = 500$	3	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*	2819.13	
		0	1.392	1.392	3.024	3.024	5	5		
$k = 750$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	5			3423.76
		0	2.134	2.134	4.634	5				
$k = 1250$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	5			4423.76
		0	2.134	2.134	4.634	5				
$k = 1500$	1	t_0^*		s_1^*	t_1	5				4903.12
		0		2.5						
$c = 2.5$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	5			3829.9
		0	1.25	2.138	3.388	5				
$c = 3.75$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	5			3900.78
		0	1.875	2.107	3.982	5				
$c = 6.25$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	5			3926.53
		0	1.875	2.107	3.982	5				
$c = 7.5$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	5			3926.53
		0	2.164	2.164	5	5				
$h_1 = 1.5$	1	t_0^*		s_1^*	t_1^*	5				3140.19
		0		1.428						
$h_1 = 2.25$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	5			3670.69
		0	1.818	2.066	3.884	5				
$h_1 = 3.75$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	5			4140.59
		0	2.164	2.164	5	5				
$h_1 = 4.5$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	5			4354.65
		0	2.164	2.164	5	5				
$h_2 = 2.5$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	5			3391.39
		0	2.164	2.164	5	5				
$h_2 = 3.75$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	5			3658.96
		0	2.164	2.164	5	5				
$h_2 = 6.25$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	5			4101.47
		0	1.538	2.100	3.639	5				
$h_2 = 7.5$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	5			4190.62
		0	1.111	2.117	3.228	5				

the constant demand rate approximation results in an overestimation (underestimation) of the total cost in the case of a decreasing (increasing) demand rate.

2. If $h_1 > h_2$ no substitution occurs.
3. Although when $h_1 < h_2$ substitution of the second product is expected, in many cases this does not happen in the presence of a high transfer cost, c .
4. The expression $t_i - s_i$ represents the time period in cycle i for the substitution of product 1 for product 2. From Table 5 it is observed that $t_i - s_i$ increases in i and this may be due to decreasing demand; the opposite is observed in Table 11. However, when a constant approximation for the demand rates is used then $t_i - s_i$ remains constant.
5. An increase in set up cost, k , leads to a decrease in the number of orders and an increase in the total cost. This is to be expected.
6. The total cost and replenishment policy seem to be more sensitive to changes in the ordering cost and the transfer cost than to changes to other costs.

Table 6

Sensitivity analysis for Example 1 assuming constant demand rates for the two products.

parameter (i)	n^*	optimal policy					Optimal cost
$k = 500$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	2956.36
		0	2.5	2.5	5	5	
$k = 750$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	3633.41
		0	2.5	2.5	5	5	
$k = 1250$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	4633.41
		0	2.5	2.5	5	5	
$k = 1500$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	5133.41
		0	2.5	2.5	5	5	
$c = 2.5$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	4014.88
		0	1.25	2.5	3.75	5	
$c = 3.75$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	4103.78
		0	1.875	2.5	4.375	5	
$c = 6.25$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	3932.54
		0	1.875	2.5	4.375	5	
$c = 7.5$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	4133.41
		0	2.5	2.5	5	5	
$h_1 = 1.5$	1	t_0^*	s_1^*	t_1^*			3472.04
		0	1.428	5			
$h_1 = 2.25$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	3847.88
		0	1.818	2.5	4.318	5	
$h_1 = 3.75$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	4370.45
		0	2.5	2.5	5	5	
$h_1 = 4.5$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	4607.5
		0	2.5	2.5	5	5	
$h_2 = 2.5$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	3540.79
		0	2.5	2.5	5	5	
$h_2 = 3.75$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	3837.1
		0	2.5	2.5	5	5	
$h_2 = 6.25$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	4315.75
		0	1.538	2.5	4.038	5	
$h_2 = 7.5$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	4396.79
		0	1.111	2.5	3.611	5	

Table 7

The optimal replenishment schedule for Example 2.

t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*
0	1.740	1.740	3.403	3.403	5	5

Table 8

The optimal replenishment schedule for Example 2 assuming constant demand for both products.

t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*
0	1.666	1.666	3.333	3.333	5	5

Table 9

The optimal replenishment schedule for Example 2 with modified data.

t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*	s_4^*	t_4^*	s_5^*	t_5^*
0	0.5	0.539	1.039	1.0671	1.567	1.585	2.085	2.094	2.593	2.593
s_6^*	t_6^*	s_7^*	t_7^*	s_8^*	t_8^*	s_9^*	t_9^*	s_{10}^*	t_{10}^*	.
3.087	3.087	3.574	3.574	4.055	4.055	4.530	4.530	5	5	.

Table 10

The optimal replenishment schedule for Example 2 with modified data (constant demand).

t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*	s_4^*	t_4^*	s_5^*	t_5^*
0	0.5	0.5	1	1	1.5	1.5	2	2	2.5	2.5
s_6^*	t_6^*	s_7^*	t_7^*	s_8^*	t_8^*	s_9^*	t_9^*	s_{10}^*	t_{10}^*	.
3	3	3.5	3.5	4	4.5	4.5	4.5	5	5	.

5. Summary and conclusions

This paper proposes a finite-planning horizon extension of the basic two-product EOQ inventory model of Drezner et al. [10]. The demand for the two products was assumed to be changing with time. Inventory policies which determine the timing, as well as the frequency, of orders were considered. When the demand rates of the two products are proportional to each other and logconcave (Assumption A1), a unique optimal replenishment schedule is shown to exist (Theorem 2). If Assumption A1 fails to hold then, for a fixed number of orders, the timing of the orders is a stationary point of the total inventory cost function but may not be unique. In a future work, it will be interesting to investigate whether this uniqueness result holds without demand proportionality. Some illustrative numerical examples were also presented to demonstrate the application of the theoretical results in different settings and lead to interesting managerial insights.

It is worth noting that the results obtained in this paper allow the treatment of some two-way substitution models. Indeed, if $h_2 > h_1$ then it can be shown that the total inventory cost of Problem P1 for model $2 \rightarrow 1$ is smaller than that of model $1 \rightarrow 2$. This implies that if Problem 2 is modified to include the permutations $1 \rightarrow 2$, $2 \rightarrow 1$ as decision variables, then $2 \rightarrow 1$ should be selected throughout the planning horizon. However, it is unlikely, in practice, for the holding and transfer costs to be equal for permutations $1 \rightarrow 2$ and $2 \rightarrow 1$ or to remain fixed over the planning horizon. Therefore, the inclusion of separate transfer costs or/and time dependent holding costs leads to a new problem, the solution of which is worth investigating.

In practice the demand for products depends largely on time. The mathematical formulation and techniques presented in this work consider arbitrary functions of time for the substitutable products, which allows the decision maker to evaluate the consequences of a wide range of policies by employing only one model. Also, since the occurrence of temporary stock-outs at retail is common for various products, the proposed model reflects a number of practical concerns with regards to product substitution and so may assist retailers in the efficient handling of inventory for these products.

Also, of particular interest is the inclusion of a penalty cost for substituting a product. This may result in the selection of a policy of full substitution in some replenishment cycles (a phenomenon which was absent in the basic model in [10]). Also, this inclusion lead to a dramatic change in the optimization problem. This problem is challenging due to the non-smoothness of the objective function.

In this paper we assumed that the replenishment rate is infinite. However, in many cases, inventory replenishment is restricted by a finite production rate. When a number of products are produced by the same machine at a finite rate, direct joint replenishment is not generally feasible. In such conditions the presence of substitution between products may mitigate the shortage effects. Also an interesting extension to the proposed model could be a model where the two products are purchased from different suppliers (are not ordered jointly in each replenishment cycle) and in addition supplier selection issues could be considered. Other possible extensions of the proposed model could include:

1. models with deteriorating items: see Bakker et al. [37],

Table 11
Sensitivity analysis for Example 2.

parameter (i)	n^*	optimal policy										Optimal cost
$k = 500$	4	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*	s_4^*	t_5^*	4510.50	
		0	1.314	1.314	2.582	2.582	3.809	3.809	5	5		
$k = 750$	4	t_0^*	s_1^*	t_1^*	t_2^*	t_2^*	s_3^*	t_3^*	s_4^*	t_4^*	5510.50	
		0	1.314	1.314	2.582	2.582	3.809	3.809	5	5		
$k = 1250$	3	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*	7110.06			
		0	1.740	1.740	3.403	3.403	5	5				
$k = 1500$	3	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*	7860.06			
		0	1.740	1.740	3.403	3.403	5	5				
$c = 2.5$	3	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*	6309.85			
		0	1.25	1.753	3.003	3.415	4.665	5				
$c = 3.75$	3	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*	6360.06			
		0	1.740	1.740	3.403	3.403	5	5				
$c = 6.25$	3	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*	6360.06			
		0	1.740	1.740	3.403	3.403	5	5				
$c = 7.5$	3	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*	6360.06			
		0	1.740	1.740	3.403	3.403	5	5				
$h_1 = 1.5$	3	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*	5626.33			
		0	1.428	1.797	3.226	3.457	4.885	5				
$h_1 = 2.25$	3	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*	6008.42			
		0	1.740	1.740	3.226	3.403	5	5				
$h_1 = 3.75$	3	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*	6711.69			
		0	1.740	1.740	3.226	3.403	5	5				
$h_1 = 4.5$	4	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*	s_4^*	t_4^*	7035.95	
		0	1.314	1.314	2.581	2.581	3.808	3.808	5	5		
$h_2 = 2.5$	3	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*	5383.3			
		0	1.740	1.740	3.226	3.403	5	5				
$h_2 = 3.75$	3	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*	5871.68			
		0	1.740	1.740	3.226	3.403	5	5				
$h_2 = 6.25$	3	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*	6838.99			
		0	1.538	1.771	3.309	3.432	4.970	5				
$h_2 = 7.5$	3	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*	7137.31			
		0	1.111	1.763	2.874	3.424	4.535	5				

- models with stock dependent items: see Urban [38],
- models with inflation or permissible delay in payment as in Gilding [39] and [30].
- models with inventory policies which not only include replenishment decisions but also pricing decisions in the presence of limited production capacity, shortages, and budget: see Ghoniem and Maddah [18] for an application of such an approach to a discrete time inventory model with substitution.

It seems that the search for the optimal policies in extensions (1) and (2) above will lead to optimization problems similar to the ones treated in this paper. However, extension (3) may not be straightforward. Proposal (4) we believe would pose a challenging problem in terms of modeling and the search for the optimal inventory policy.

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Appendix A. Proof of Theorem 1

Assume that the optimal solution in cycle i is given by $(\hat{t}_{i-1}, \tilde{s}_i, \hat{t}_i)$, where $\hat{t}_{i-1} \leq \tilde{s}_i < \hat{t}_i$, and $\tilde{s}_i \neq \hat{s}_i$. Define the function

$$J(s) := \int_{\hat{t}_{i-1}}^{\hat{t}_i} \{h_1 D_1(t) + h_2 D_2(t)\} (t - \hat{t}_{i-1}) dt + \int_s^{\hat{t}_i} \{c - (h_2 - h_1)(t - \hat{t}_{i-1})\} D_2(t) dt.$$

Table 12
Sensitivity analysis for Example 2 assuming constant demand rates for the two products.

parameter (i)	n^*	optimal policy										Optimal cost
$k = 500$	4	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*	s_4^*	t_4^*	4485.94	
		0	1.25	1.25	2.5	2.5	3.75	3.75	5	5		
$k = 750$	4	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*	s_4^*	t_4^*	5485.94	
		0	1.25	1.25	2.5	2.5	3.75	3.75	5	5		
$k = 1250$	3	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*			7064.58	
		0	1.666	1.666	3.333	3.333	5	5				
$k = 1500$	3	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*			7814.58	
		0	1.666	1.666	3.333	3.333	5	5				
$c = 2.5$	3	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*			6266.41	
		0	1.25	1.666	2.916	3.333	4.583	5				
$c = 3.75$	3	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*			6314.58	
		0	1.666	1.666	3.333	3.333	5	5				
$c = 6.25$	3	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*			6314.58	
		0	1.666	1.666	3.333	3.333	5	5				
$c = 7.5$	3	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*			6314.58	
		0	1.666	1.666	3.333	3.333	5	5				
$h_1 = 1.5$	2	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*					5559.6	
		0	1.428	2.5	3.928	5						
$h_1 = 2.25$	3	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*			5967.71	
		0	1.666	1.666	3.333	3.333	5	5				
$h_1 = 3.75$	3	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*			6661.46	
		0	1.666	1.666	3.333	3.333	5	5				
$h_1 = 4.5$	4	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*	s_4^*	t_4^*	7006.25	
		0	1.25	1.25	2.5	2.5	3.75	3.75	5	5		
$h_2 = 2.5$	3	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*			5351.04	
		0	1.666	1.666	3.333	3.333	5	5				
$h_2 = 3.75$	3	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*			5832.81	
		0	1.666	1.666	3.333	3.333	5	5				
$h_2 = 6.25$	3	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*			6788.94	
		0	1.538	1.666	3.205	3.333	4.871	5				
$h_2 = 7.5$	3	t_0^*	s_1^*	t_1^*	s_2^*	t_2^*	s_3^*	t_3^*			7085.42	
		0	1.111	1.666	2.777	3.333	4.444	5				

It follows that

$$J'(s) = -\{c - (h_2 - h_1)(s - \hat{t}_{i-1})\}D_2(s).$$

It is clear that $J'(s) \leq 0$, if $\gamma \leq 0$, or if $\gamma \geq \hat{t}_i - \hat{t}_{i-1}$ with unique minimum for $J(s)$ at $s = \hat{t}_i$. Also, the minimum is unique if $\hat{t}_{i-1} < \hat{t}_{i-1} + \gamma \leq \hat{t}_i$ and occurs at $s = \hat{t}_{i-1} + \gamma$. Therefore, the function V has a unique minimum in $[\hat{t}_{i-1}, \hat{t}_i]$ given by $\hat{s}$: see (15), in other words $V(\hat{t}_{i-1}, \hat{s}_i, \hat{t}_i) < V(\hat{t}_{i-1}, \tilde{s}_i, \hat{t}_i)$. This leads to a contradiction with the the assertion that $(\hat{t}_{i-1}, \tilde{s}_i, \hat{t}_i)$ is optimal.

Appendix B. Proof of Lemma 1

The functions R_1 is continuously differentiable in $\tilde{\Omega}$. Also, R_2 is differentiable in the space $\{(x, y): x < y \leq T, \text{ and } x + \gamma < y\}$, and the space $\{(x, y): x < y \leq T, \text{ and } x + \gamma > y\}$. Therefore it suffices to check that R_2 is differentiable on the line segment $(x, x + \gamma)$. Computation, using (19), shows that

$$\begin{aligned}\partial_x r_2(x + \gamma, y) &= - \int_{x+\gamma}^y D_2(t) dt, \\ \partial_y r_2(x + \gamma, y) &= \{y - (x + \gamma)\} D_2(y).\end{aligned}$$

Clearly $\partial_x r_2(x + \gamma, x + \gamma) = 0$, and $\partial_y r_2(x + \gamma, x + \gamma) = 0$. This leads, by (21) to the required result.

Appendix C. Proof of Lemma 2

We shall only show the lemma for $j = 2$. The proof for $j = 1$ is similar.

Recall (36), (37), (19) and (24)–(28) to get

$$\begin{aligned}\mathcal{L}_x^{(2)} r_2(x, y) &= D_2(x) - D_2(y) + \frac{f_2'(x)}{f_2(x)} \int_x^y D_2(t) dt \\ &= \int_x^y \{f_2(t) - f_2(x)\} D_2(t) dt,\end{aligned}\quad (C1)$$

and

$$\mathcal{L}_x^{(2)} r_2(x + \gamma, y) = \int_{x+\gamma}^y \{f_2(t) - f_2(x)\} D_2(t) dt. \quad (C2)$$

Also, direct algebra shows that $\mathcal{L}_y^{(2)} r_2(x, y) = 0$. Recall the assumption that f_2 is non-decreasing. The proof is then immediate by using the definition of R_2 in (21).

Appendix D. Proof of Theorem 2

The proof is similar in spirit to that found in [30]. We shall only include details up to a key step related to Assumption A1.

The proof is based on the following induction hypothesis:

The Induction Hypothesis:

There exist functions $\{\tau_j\}$, $j = 0, \dots, n-1$ which are piecewise continuously differentiable such that $\tau_0 \equiv 0$,

$$\tau_j(0) = 0, \quad (D1)$$

and

$$0 < \tau_j'(\eta^\pm) < 1 \quad \text{for } 0 < \eta \leq T \quad (D2)$$

and $j = 1, 2, \dots, n-1$ with the following properties. For every $0 < h \leq T$, the minimum of G_n under the constraint

$$0 = t_0 < t_1 < \dots < t_n = h \quad (D3)$$

exists, is unique, and is given by

$$t_j = \tau_j(t_{j+1}) \quad \text{for } j = n-1, n-2, \dots, 1, 0.$$

Furthermore, the minimum value $g_n(h)$ satisfies

$$g_n'(h) = (\partial_y R)(\tau_{n-1}(h), h). \quad (D4)$$

End of the induction hypothesis

Let $n = 1$, then (42) and (23) give $G_n(t_0, t_1) = R(0, h)$, and $g_n(h) = R(0, h)$. It is clear that the theorem is true in this case. Assume now that the theorem is true for $n \geq 1$, and consider the problem of minimizing $G_{n+1}(t_0, \dots, t_{n+1})$ with $t_0 = 0 < t_1 < \dots < t_{n+1} = h \leq T$.

The Dynamic Programming principle of Bellman leads to:

$$\sigma_{n+1}(\eta, h) = g_n(\eta) + R(\eta, h). \quad (D5)$$

We aim to minimize σ_{n+1} under the constraint $0 < \eta < h$. This function is continuous on the interval $[0, h]$. Also,

$$\begin{aligned}(\partial_x \sigma_{n+1})(\eta, h) &= g_n'(\eta) + (\partial_x R)(\eta, h) \\ &= (\partial_y R)(\tau_{n-1}(\eta), \eta) + (\partial_x R)(\eta, h).\end{aligned}\quad (D6)$$

This is justified by (D4) in the induction hypothesis.

Now, again by (D1), in the induction hypothesis and the definition of R

$$\begin{aligned}(\partial_x \sigma_{n+1})(0, h) &= (\partial_y R)(\tau_{n-1}(0), 0) + (\partial_x R)(0, h) \\ &= (\partial_y R)(0, 0) + (\partial_x R)(0, h) \\ &= \sum_{j=1}^2 \{(\partial_y R_j)(0, 0) + (\partial_x R_j)(0, h)\}.\end{aligned}$$

Using the definitions of R_1 and R_2 in (20) and (21) respectively together with (24) and (25) we assert that $(\partial_x \sigma_{n+1})(0, 0) < 0$. Likewise, we get

$$\begin{aligned} (\partial_x \sigma_{n+1})(h, h) &= (\partial_y R)(\tau_{n-1}(h), h) + (\partial_x R)(h, h) \\ &= \sum_{j=1}^2 \{(\partial_y R_j)(\tau_{n-1}(h), h) + (\partial_x R_j)(h, h)\} > 0. \end{aligned}$$

It follows from continuity that there exist at least one $\eta \in (0, h)$ such that

$$(\partial_x \sigma_{n+1})(\eta, h) = 0. \quad (D7)$$

Note by Remark 2 that we may differentiate $(\partial_x \sigma_{n+1})(\eta, h)$ with respect to η and h except at the points $\tau'_{n-1}(\eta^+) \neq \tau'_{n-1}(\eta^-)$, $\tau_{n-1}(\eta) + \gamma = \eta$, and $\eta + \gamma = h$. Denote such points η , where $\tau'_{n-1}(\eta^+) \neq \tau'_{n-1}(\eta^-)$, by ξ_k , $k = 1, \dots, m$. Also, note that, by the induction hypothesis, there exists at most one point where $\tau_{n-1}(\eta) + \gamma = \eta$. Denote this point by ξ_0 . Excluding all points $\eta = \xi_k$, $k = 0, \dots, m$, and $\eta + \gamma = h$, we differentiate $(\partial_x \sigma_{n+1})(\eta, h)$ with respect to η to get

$$(\partial_x^2 \sigma_{n+1})(\eta, h) = (\partial_x \partial_y R)(\tau_{n-1}(\eta), \eta) \tau'_{n-1}(\eta) + (\partial_y^2 R)(\tau_{n-1}(\eta), \eta) + (\partial_x^2 R)(\eta, h). \quad (D8)$$

Using the definition of R , $(\mathcal{L}_x^{(j)}, \mathcal{L}_y^{(j)}, j = 1, 2)$, (39), and (40), we may write (D8) as

$$\begin{aligned} (\partial_x^2 \sigma_{n+1})(\eta, h) &= -\{1 - \tau'_{n-1}(\eta)\}(\partial_x \partial_y R)(\tau_{n-1}(\eta), \eta) + \{(\mathcal{L}_x R)(\eta, h) + (\mathcal{L}_y R)(\tau_{n-1}(\eta), \eta)\} \\ &\quad - f(\eta)(\partial_x \sigma_{n+1})(\eta, h) - (\partial_x \partial_y R)(\eta, h). \end{aligned} \quad (D9)$$

Under assumptions (A1) and (A2) and invoking the induction hypothesis, Lemma 2, and (D7)

$$(\partial_x^2 \sigma_{n+1})(\eta, h) \geq -(\partial_x \partial_y R)(\eta, h) > 0. \quad (D10)$$

Also, letting $\eta \rightarrow \xi_k^\pm$, and $\eta = h - \gamma$, separately, we have

$$(\partial_x^2 \sigma_{n+1})(\eta^\pm, h) \geq -(\partial_x \partial_y R)(\eta^\pm, h) > 0. \quad (D11)$$

Therefore, (D7) has a unique solution. This solution is the global minimum of $\sigma_{n+1}(\cdot, h)$ for every $h \in (0, T]$.

The rest of the proof follows the steps as in [30] and is omitted.

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